PRODUCT USER MANUAL

For Global Ocean Biogeochemistry Multiyear Product

GLOBAL_MULTIYEAR_BGC_001_029
Issue: 2.3

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/01	All	initial version	C. Perruche	Y Drillet
1.1	2019/03/19		Extension Addition of variables ph and spco2 in monthly files	C. Perruche	Y. Drillet
2.0	2022/11		New template	A. Biardeau	Copernicus Marine Product Management
2.1	2023/06/16		Extension Add interim dataset Renamed datasets	C. Perruche	Copernicus Marine Product Management
2.2	2024/05/31		Interim dataset at Month -2	C. Perruche	O. Le Galloudec
2.3	2025/05/30		Merge dataset myint and my	C. Perruche	O. Le Galloudec

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus MMarine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS global biogeochemical multi-year hindcast (no data assimilation) GLOBAL_MULTIYEAR_BGC_001_029 : it provides with daily and monthly fields from 1993 to M-2.

This product is global. It is defined on a standard grid at 1/4 degree (approx. 25km) and on 75 standard levels. It is interpolated from the 1/12 degree and 50 vertical levels Arakawa C native grid. All variables are on the same grid points.

GLOBAL_MULTIYEAR_BGC_001_029 product is organised in three datasets:

- **cmems_mod_glo_bgc_my_0.25deg_P1D-m** which contains 3D daily mean fields of chlorophyll, nitrate, phosphate, silicate, dissolved oxygen, net primary production.

- **cmems_mod_glo_bgc_my_0.25deg_P1M_m** which contains 3D monthly mean fields of chlorophyll, nitrate, phosphate, silicate, dissolved oxygen, net primary production, iron, phytoplankton concentration in carbon, pH, and 2D monthly mean fields of surface partial pressure of CO2.

- **cmems_mod_glo_bgc_my_0.25deg_static** which contains the static fields for the system: coordinates, mean sea surface level, mask and bathymetry.

The product is published on the Copernicus Marine dissemination server after automatic and human quality controls. Files downloaded are in NetCDF4 format and follow CF-1.6 convention.

This multi-year product is described in the Quality Information Document (QUID – see in REFERENCES)

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

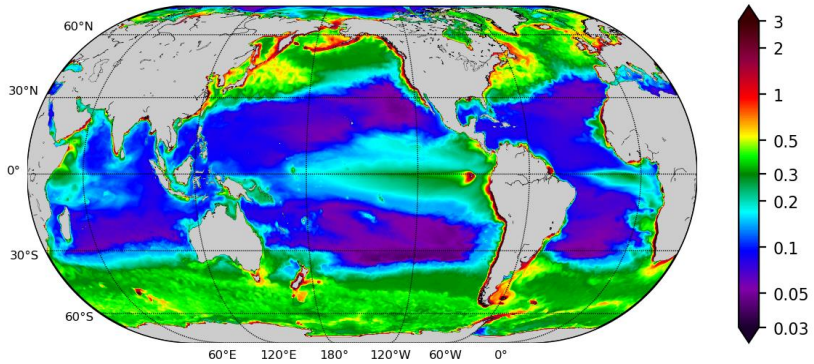
a) General Information

Product Lines	GLOBAL_MULTIYEAR_BGC_001_029
Geographical coverage	Global : 180°W-180°E ; 89°S – 90°N
Variables	Chlorophyll Nitrate Phosphate Silicate Dissolved Oxygen Net Primary Production Iron Phytoplankton Concentration in Carbon pH Surface partial pressure of CO2
Product Type	Reanalysis
Available time series	From 1993/01/01 to M-2
Temporal resolution	Daily - Monthly means
Update Frequency	Monthly
Target delivery time	N/A
Horizontal resolution	1/4 degree
Number of vertical levels	75
Format	NetCDF4

b) Production System Description

The Mercator-Ocean global ocean biogeochemical multi-year product at 1/4 degree provides daily and monthly mean files of chlorophyll, nitrate, phosphate, silicate, oxygen and primary production (iron and phytoplankton concentration in addition in monthly files).

GLOBAL (180°W-180°E ; 89°S – 90°N)
1/4°; regular grid ; 1440 x 681
This product is global with dedicated projection and spatial resolution. It is defined on a standard collocated grid at 1/4 degree (approx. 25 km). The parameters are interpolated from the native grid model, the 1/4 degree and 75 vertical levels Arakawa C native grid.



Vertical grid	75 vertical levels
System Name	BIORYS4v4
Model version	NEMO3.6 – PISCESv2
Horizontal Resolution	1/4°
Vertical Resolution	75 Z levels
Ocean dynamical forcings	Mercator simulation FREEGLORYS2V4: daily means
Bathymetry	ETOPO1 for the deep ocean and GEBCO1 close to the cost and slope
Initial conditions	World Ocean Atlas 2013 for no3, po4, si, o2 GLODAPv2 for Alkalinity et Dissolved Inorganic Carbon Model outputs (climatological run) for iron and Dissolved Organic Carbon
Atmospheric forcings	ERA-Interim: daily means
External sources of nutrients	Atmospheric deposition (dust) of nitrate, phosphate, silicate and iron (monthly climatology) Rivers (monthly climatology): Rivers are colocalized with runoffs of the physical model. Inputs from sediments: Inputs of Iron are parameterized with the variance of the bathymetry

	Atmospheric CO2: Monthly global means of atmospheric CO2 value varying with anthropogenic emissions
Data Assimilation scheme	N/A
Assimilated observations	N/A

c) Details of the datasets

Product Lines	GLOBAL_MULTIYEAR_BGC_001_029
Geographical coverage	Global
Datasets: **cmems_mod_glo_bgc_my_0.25deg_P1D-m** **cmems_mod_glo_bgc_my_0.25deg_P1M-m** **cmems_mod_glo_bgc_my_0.25deg_static**	
Product Type	Renalysis
Update frequency	Yearly
Available time series	1993/01/01 to M-2
Target delivery time	N/A
Temporal resolution	Daily mean Monthly mean

Dataset : cmems_mod_glo_bgc_my_0.25deg_P1D-m contains 3D daily mean fields of chlorophyll, nitrate, phosphate, silicate, dissolved oxygen, net primary production
chl [mg.m-3] Chlorophyll concentration mass_concentration_of_chlorophyll_a_in_sea_water
no3 [mmol.m-3] Nitrate concentration mole_concentration_of_nitrate_in_sea_water
po4 [mmol.m-3] Phosphate concentration mole_concentration_of_phosphate_in_sea_water
si [mmol.m-3] Silicate concentration mole_concentration_of_silicate_in_sea_water
nppv [mg.m-3.day-1] Net primary production net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
o2 [mmol.m-3] Dissolved oxygen mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
Dataset : cmems_mod_glo_bgc_my_0.25deg_P1M-m

contains 3D monthly mean fields of chlorophyll, nitrate, phosphate, silicate, dissolved oxygen, net primary production, iron, phytoplankton concentration in carbon and pH. contains 2D monthly mean fields of surface partial pressure of CO2
chl [mg.m-3] Chlorophyll concentration mass_concentration_of_chlorophyll_a_in_sea_water
no3 [mmol.m-3] Nitrate concentration mole_concentration_of_nitrate_in_sea_water
po4 [mmol.m-3] Phosphate concentration mole_concentration_of_phosphate_in_sea_water
si [mmol.m-3] Silicate concentration mole_concentration_of_silicate_in_sea_water
nppv [mg.m-3.day-1] Net primary production net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
o2 [mmol.m-3] Dissolved oxygen mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
fe [mmol.m-3] Dissolved iron concentration mole_concentration_of_silicate_in_sea_water
phyc [mmol.m-3] Phytoplankton concentration in carbon mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
ph [1] pH sea_water_ph_reported_on_total_scale
spco2 [Pa] surface partial pressure of CO_2 surface_partial_pressure_of_carbon_dioxide_in_sea_water
Dataset: cmems_mod_glo_bgc_my_0.25deg_static contains the static fields for the system: coordinates, mask.
e1t [m] Cell dimension along X axis
e2t [m] Cell dimension along Y axis
e3t [m] Cell dimension along Z axis cell_thickness
mask [1] Land-sea mask: 1 = sea ; 0 = land sea_binary_mask
deptho [m] Bathymetry

sea_floor_depth_below_geoid
deptho_lev [1] Model level number at sea floor model_level_number_at_sea_floor

d) Processing information

The multiyear dataset is homogeneous along the whole period of time. lit is extended every month and is homogeneous with the multiyear dataset.

e) Time averaging

For the monthly dataset, the fields are monthly means over the calendar month (first to last day of the month).
For the daily dataset, the fields are daily means over a day (midnight to midnight, centered at noon).

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF
To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_mod_glo_bgc_my_0.25deg_P1D-m	mercatorfreebiorys2v4_global_mean_${YYYYMMDD}.nc	1.1
cmems_mod_glo_bgc_my_0.25deg_P1D-m	mercatorfreebiorys2v4_global_mean_${YYYYMM}.nc	0.7
cmems_mod_glo_bgc_my_0.25deg_static	GLO-MFC_001_029_${field}.nc	0.2

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in ANNEX.

7) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-029.pdf

8) ANNEX

```
netcdf mercatorfreebiorys2v4_global_mean_20201231 {
dimensions:
        longitude = 1440 ;
        latitude = 681 ;
        depth = 75 ;
        time = UNLIMITED ; // (1 currently)
variables:
        float longitude(longitude) ;
                longitude:valid_min = -180.f ;
                longitude:valid_max = 179.75f ;
                longitude:step = 0.25f ;
                longitude:units = "degrees_east" ;
                longitude:unit_long = "Degrees East" ;
                longitude:long_name = "Longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:axis = "X" ;
        float latitude(latitude) ;
                latitude:valid_min = -80.f ;
                latitude:valid_max = 90.f ;
                latitude:step = 0.25f ;
                latitude:units = "degrees_north" ;
                latitude:unit_long = "Degrees North" ;
                latitude:long_name = "Latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:axis = "Y" ;
        float depth(depth) ;
                depth:valid_min = 0.50576f ;
                depth:valid_max = 5902.058f ;
                depth:units = "m" ;
                depth:positive = "down" ;
                depth:unit_long = "Meters" ;
                depth:long_name = "Depth" ;
                depth:standard_name = "depth" ;
                depth:axis = "Z" ;
        float time(time) ;
                time:long_name = "Time (hours since 1950-01-01)" ;
                time:standard_name = "time" ;
                time:calendar = "gregorian" ;
                time:units = "hours since 1950-01-01 00:00:00" ;
                time:axis = "T" ;
        float chl(time, depth, latitude, longitude) ;
                chl:long_name = "Total Chlorophyll" ;
                chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
                chl:units = "mg m-3" ;
                chl:unit_long = "milligram of Chlorophyll per cubic meter" ;
                chl:_FillValue = 9.96921e+36f ;
                chl:valid_min = 0.002648598f ;
                chl:valid_max = 14.42348f ;
        float no3(time, depth, latitude, longitude) ;
                no3:long_name = "Nitrate" ;
```

```
                no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
                no3:units = "mmol m-3" ;
                no3:unit_long = "millimoles of Nitrate per cubic meter" ;
                no3:_FillValue = 9.96921e+36f ;
                no3:valid_min = 0.f ;
                no3:valid_max = 73.22569f ;
        float nppv(time, depth, latitude, longitude) ;
                nppv:long_name = "Total Primary Production of Phyto" ;
                nppv:standard_name =
"net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water"
;
                nppv:units = "mg m-3 day-1" ;
                nppv:unit_long = "milligrams of Carbon per cubic meter per day" ;
                nppv:_FillValue = 9.96921e+36f ;
                nppv:valid_min = 0.f ;
                nppv:valid_max = 534.4241f ;
        float o2(time, depth, latitude, longitude) ;
                o2:long_name = "Dissolved Oxygen" ;
                o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
                o2:units = "mmol m-3" ;
                o2:unit_long = "millimoles of Oxygen per cubic meter" ;
                o2:_FillValue = 9.96921e+36f ;
                o2:valid_min = 0.2499826f ;
                o2:valid_max = 529.7874f ;
        float po4(time, depth, latitude, longitude) ;
                po4:long_name = "Phosphate" ;
                po4:standard_name = "mole_concentration_of_phosphate_in_sea_water" ;
                po4:units = "mmol m-3" ;
                po4:unit_long = "millimoles of Phosphate per cubic meter" ;
                po4:_FillValue = 9.96921e+36f ;
                po4:valid_min = 3.505161e-05f ;
                po4:valid_max = 8.512593f ;
        float si(time, depth, latitude, longitude) ;
                si:long_name = "Dissolved Silicate" ;
                si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
                si:units = "mmol m-3" ;
                si:unit_long = "millimoles of Silicate per cubic meter" ;
                si:_FillValue = 9.96921e+36f ;
                si:valid_min = 0.2690167f ;
                si:valid_max = 770.1432f ;

// global attributes:
                :product = "GLOBAL_REANALYSIS_BIO_001_029" ;
                :producer = "CMEMS - Global Monitoring and Forecasting Centre" ;
                :title = "Daily mean fields for product GLOBAL_REANALYSIS_BIO_001_029" ;
                :area = "GLOBAL" ;
                :quality_information_document =
"http://marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-029.pdf" ;
                :Conventions = "CF-1.6" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :references = "http://marine.copernicus.eu" ;
                :source = "MERCATOR FREEBIORYS2V4" ;
```

```
                :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-
and-licence/" ;
                :dataset = "global-reanalysis-bio-001-029-daily" ;
                :institution = "Mercator Ocean" ;
                :product_user_manual =
"http://marine.copernicus.eu/documents/PUM/CMEMS-GLO-PUM-001-029.pdf" ;
                :_NCProperties = "version=2,netcdf=4.7.1,hdf5=1.8.18," ;
}
```